MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION - OCTOBER 2025
THIRD SEMESTER
Discipline Specific Elective (DSE)-MG3DSEBOT201 – HERBAL TECHNOLOGY
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Part A
Very Short answer type questions

Answer any 10 questions Each question carries 2 marks [2x10 = 20]

1. **Name any four plants maintained in the home gardens of Kerala.**
   [K/CO1]
   Coconut, banana, drumstick, turmeric, ginger, tulsi, papaya, etc. – **any four – 2 marks.**
2. **Mention the advantages of microwave assisted extraction of phytochemicals.**
   [K/CO4]
   - Faster than traditional methods (needed only minutes vs. hours)
   - Less solvent required (eco-friendly)
   - Higher yields due to better cell disruption
   - Energy-efficient
   - Good for heat-sensitive compounds (if conditions are controlled) – **Any two points -2 marks**
3. **Name a commonly used herbal plant and mention its therapeutic application as home remedy.**
   [K/CO2]
   Turmeric, *Aloe vera*, pepper, chilly, tulsi, etc. – **anyone with one therapeutic use – 2 marks**
4. **How is MBG involved in research on medicinal and aromatic plants?**
   [U/CO1,2]
   Malabar Botanical Garden –involved in: Conservation of Aquatic and Wetland Plants, Germplasm Collection and Genetic Resources, Taxonomy & Biodiversity Documentation, Education, Capacity Building & Outreach programmes – **any two thrust areas of MBG -2 marks**
5. **What is phytochemistry?**
   [U/CO1]
   Chemistry of plants **-** branch of science that deals with the study of the chemical compounds produced by plants**. – 2 marks**
6. **Mention any two advantages of herbal dyes.**
   [U/CO4]
   Eco-friendly, biodegradable, non-toxic, skin-friendly, renewable, sustainable, medicinal, etc. – **any 2 points- 2 marks.**
7. **What is swarming?**
   [K/CO4]
   Out of syllabus – **Give full marks if attended**
8. **What is the fundamental principle on which solvent extraction is based?**
   [U/CO4]

   Solvent extraction is a method to separate compounds based on their relative solubilities in two different immiscible liquids, usually water (polar) and an organic solvent (non-polar or semi-polar).

9. **Name an equipment used in extraction procedure.**
   [U/CO1,2]
   Mortar and pestle, Soxhlet apparatus, pulverizer, Microwave extractor, separating funnel, etc. – **anyone – 2 marks**.

10. **Name any two herbal gardens maintained by research centres or educational institutions**. [K/CO1]

    JNTBGRI, MBGIPS, etc. – **any two – 2 marks**

11. **Judge the impact of use of synthetic chemicals in herbal plant cultivation.**

    [U/CO2]

    Reduced medicinal value, water and soil pollution, or any other impact of chemical fertiliser – **liberal valuation – 2 marks**

12. **Define transesterification.**

    [U/CO4]

    A stage in biodiesel production **or** conversion of triglycerides into fatty acid methyl esters.- **liberal valuation.**

Part B

Short Essay type questions, Answer any 8 questions, **Each question carries 5 marks** [5 x 8 = 40]

13. **Discus the sources of herbs.**

    [U/CO1]

    Nurseries, herbal gardens, wild sources, micropropagation labs etc. Question is not unambiguous/ Out of syllabus – **full marks if attended**

14. **Comment on the characters of a flower to attract honey bees for pollination.**

    [U/CO4]

    Bright colours, fragrance, nectar, diurnal or day-blooming nature, etc. – **liberal valuation**

15. **Explain how adding soap to tobacco decoction improves its effectiveness as a pesticide?** [U/CO4]

    Allows the liquid to spread evenly over the leaves, improves adhesion of nicotine to plant surface and cause effective penetration. – **Give full marks for any two relevant points**.

16. **Explain the occurrence, properties and uses of secondary metabolites in medicine.** [K/CO3,5]

    Question is not unambiguous.

    Secondary plant metabolites are chemical/organic compounds produced by the plant cell through metabolic pathways derived from the primary metabolic pathways.

    These compounds are not directly involved in the growth or reproduction in plants.

    But they play a crucial role in plant's defence mechanism, pigmentation and human medicine.

    Or Any two secondary metabolites and their 2 medicinal uses – Give full mark.

17. **Explain applications of herbal dyes**. [K/CO4]

    Herbal dyes can be applied in different fields such as textile, cosmetic, food, hair treatment, pharmaceutical, handicraft and art, leather and paper industry. – **Explain any three points with example – Full mark.**

18. **Explain the role of home gardens and botanical gardens in the conservation of herbal plants.**

    [U/CO2]

    Biodiversity Conservation through home and botanical gardens:

- Herbal gardens act as a repository for medicinal and aromatic plants, helping to conserve plant diversity.
- They play a vital role in protecting rare, endangered, and threatened (RET) species.

Botanical gardens are important in Scientific Study: Herbal gardens provide a controlled environment for researchers to study the

- Medicinal, nutraceutical and other properties of herbals,
- their active compounds, and
- potential applications in various fields.

Germplasm Conservation: They act as ex-situ conservation centres for herbals.

- It ensures their availability for future research and development. – **Any 5 relevant points- Full mark.**

19. **Explain the properties of supercritical fluids used in the extraction of secondary metabolites**.

[U/CO4]

A supercritical fluid is a state of matter that occurs when a substance is above its critical temperature and critical pressure.

It is the point at which the distinction between liquid and gas phases disappears.

- It has no distinct liquid or gas phase
- But exhibits properties of both liquids and gases
    - Like a gas, it can diffuse easily and fill a container
    - Like a liquid, it can dissolve substances (like solutes or oils)

20. **Clarify the role of soap solution in neem decoction formulation.**

[U/CO4]

Allows the liquid to spread evenly over the leaves, improves adhesion of azadiractine to plant surface and cause effective penetration. – **Give full marks for any two relevant points**.

21. **How does herbal technology promote eco-friendly and sustainable healthcare practices**? [K/CO1]

- Products of herbal technology are generally more eco-friendly compared to synthetic products.
- For example, Growing medicinal plants (Herbals) consumes less chemical input and generates a lower carbon footprint (release of green house gases).
- Sustainable harvesting practices also conserve biodiversity.
- Support for herbal products may thus help a healthy planet.
- Choosing herbal products from ethical sources promotes conservation efforts and reduces harm to the environment.

22. **Analyse the multifaceted role of Malabar Botanical Garden as a centre of conservation, research and education.**

[An/CO2]

The major thrust areas of the MBGIPS shows the evidence of its multifaceted role.

a. Conservation of Aquatic and Wetland Plants,Cryptogams, Bryophytes, Pteridophytes and angiosperms.
b. Biodiversity Documentation: Surveying, exploring fields, documenting species distributions, collecting specimens, identifying,cataloguing and depositing specimens in their herbarium (MBGH) is an important research focus.
c. Horticulture, Ornamental Plant Development and Breeding: Development of ornamental varieties, especially in aquatic plants like Nymphaea (water lilies) andNelumbo (lotus), with breeding programmes.
d. Education, Capacity Building & Outreach: Academic attachment/programmes for students at UG/PG level; internships; training and educationin plant sciences.

Part C

Essay type questions: **Answer any 1 question,** Question carries 10 marks; (1x10: 10)

23. **Discuss the advantages and limitations of using alcohol and acetone in plant extraction methods, with suitable examples.**
[K/CO4]

Extraction can be defined as the separation of desired active compound/s from plant tissues using specific solvents - Solvent extraction - Two common solvents used in herbal extraction are – alcohol and acetone.

a) Alcohol - two common alcohols include ethanol and methanol.

Advantages - Polarity: Polar

Boiling Point: Ethanol ~78°C, Methanol ~65°C

Solubility: Soluble in water

Extracts a wide range of polar and semi-polar compounds: alkaloids, flavonoids, glycosides, tannins, saponins, and phenolic compounds.

Ethanol is food- and pharma-safe, commonly used in tinctures and liquid extracts.

Methanol is more polar and often used in analytical research.

Limitations: extracts are toxic for consumption, selective solubility, loss of heat-sensitive compounds, high purity grade ethanol is expensive, etc.

b) Acetone - a colourless, volatile, and flammable liquid.

Advantages:

- Polarity: Medium-polar (semi-polar)
- Boiling Point: ~56°C
- Solubility: Miscible with water and many organic solvents.
- Efficient in extracting flavonoids, terpenoids, and some alkaloids.
- Volatile and evaporates easily, which aids in drying the extract.
- Used mainly in lab-scale or preparative extractions.

Limitations: highly flammable and toxic if inhaled or ingested.

- Highly polar or non-polar phytochemicals (like sugars, proteins, and lipids) may not be efficiently extracted.
- Residual solvent problem
- Evaporation and loss of volatiles.

24. **Describe biopesticides and explain their significance.**
[U/CO4]

Biopesticides are natural substances derived from living organisms (such as plants, microorganisms, etc.) that are used to control pests, weeds, and plant diseases.

- They are eco-friendly, biodegradable, and less harmful to humans and non-target organisms.
  Examples:
  1. Neem decoction and
  2. Tobacco decoction
- Acts as a natural insect repellent against pests like aphids, whiteflies, and caterpillars.
- Controls fungal infections such as powdery mildew and rust in crops.
- Used for protecting seeds from fungal and bacterial infections before sowing.
- Reduces nematode population and improves soil health.
- Sprayed indoors or used for cleaning to repel mosquitoes and insects.
- Added to bathwater or floor wash for natural disinfection.
- Used as a rinse to treat dandruff, lice, and scalp infections. – **Give marks for significance/applications.**